

Part-C

10 each

5. Define the following with the help of examples :

(i) Circular linked list

(ii) Inverted linked list

6. Differentiate between linear and doubly linked list.

What are the advantages of using linked lists over array ?

Part-D

10 each

7. Write an algorithm to insert a new node at the end of the stack.

8. Explain different operations that may be performed on queues with the help of example.

Part-E

10 each

9. Write a pseudo algorithm for inorder traversal of a binary tree. Explain with the help of an example.

10. Write a pseudo algorithm for Linear search.

Roll No.

Total No. of Questions : 10] [Total No. of Printed Pages : 4
(1048)

**B.C.A. (CBCS) RUSA IInd Semester
Examination**

4030

DATA STRUCTURES

Paper : BCA-0204

Time : 3 Hours]

[Maximum Marks : 70

Note :- Part-A is compulsory. Candidates need to attempt
one question from each Parts B, C, D and E.

Part-A

(Compulsory Question)

1. Answer the following objective types questions :

- (a) The end at which new elements are added in a queue is called
- (b) The the number of children a tree has.

C-662

(1)

Turn Over

- (c) Stacks work on principles.
- (d) An example of dynamic data structures is
- (e) is the procedure by which each node in the tree is visited once in a systematic manner.
- (f) The complexity of quick sort algorithm is
- (g) Left, right, root traversal is known as traversal.
- (h) is a finite ordered set of homogeneous elements.
- (i) data structures are which expand or shrink during program execution and their associated memory location change.
- (j) is the process of combining records in two different sorted files into a single sorted file.

10×1=10

2. Answer the following questions in short :

- (a) What do you mean by Time and Space Complexity ? Explain.
- (b) What is a binary threaded tree ? Explain.
- (c) What are push and pop operations on stack ?
- (d) Differentiate with examples between static and dynamic data structures.
- (e) Differentiate between LIFO and FIFO. $5 \times 4 = 20$

Part-B

10 each

3. What do you understand by the following operations on data structures ?

- (i) Merging
- (ii) Searching
- (iii) Sorting
- (iv) Traversing

4. How arrays are arranged in Memory ? Differentiate between one dimensional and multidimensional array.